Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> The power dissipated in the headlightbulb can be calculated using the formula: $P = \frac{V^2}{R}$, where P is the power, V is the voltage, and R is the resistance. Given that the power P is 40 watts and the voltage V is 12 volts, we can solve for the resistance R:

$$40 = \frac{12^2}{R}$$

$$40 = \frac{144}{R}$$

$$R = \frac{144}{40}$$

$$R = 3.6 \, \Omega$$

Next, we use Ohm's Law to find the current I:

$$I = \frac{V}{R}$$

$$I = \frac{12}{3.6}$$

$$I = 3.33 \, \text{amps}$$

So, the headlightbulb will draw 3.33 amps and its resistance must be 3.6 ohms. </think>
<answer>E</answer>